

Revision Notes for Class 8 Maths Chapter 7 Area

Understanding the concept of area is essential in geometry, especially when dealing with rectangles, squares, triangles, and other polygons. In everyday life, we often come across questions such as how much space a rectangle covers or how different shapes can have the same perimeter but different areas. Knowing how to calculate the area helps compare and analyze such situations efficiently.

Area of Rectangles and Squares

To calculate the area of a rectangle, simply multiply its length by its width. For example, a rectangle of 7 cm by 4 cm has an area of 28 cm^2 , while one with sides 8 cm and 3 cm has an area of 24 cm^2 . The rectangle that covers more unit squares (i.e., has a bigger product of sides) will require more rangoli powder to fill, and thus, has a greater area. The area of a square, being a special case of a rectangle, is given by $\text{side} \times \text{side}$.

Comparing Perimeter and Area

Perimeter alone cannot measure area. Two regions might have the same perimeter but very different areas. For example, you might find two rectangles—one long and thin, the other more square-shaped—having equal perimeters but different areas. Sometimes, a shape with a greater perimeter can even have a smaller area.

Triangles in Rectangles

When a rectangle is divided along its diagonal, it forms two identical triangles. Each triangle's area is exactly half that of the rectangle. For a rectangle measuring 7 cm by 4 cm, each triangle has an area of $\frac{1}{2} \times 7 \times 4 = 14 \text{ cm}^2$. This idea helps reinforce why the area of a triangle is given by one half the product of its base and corresponding height.

Area of a Triangle

The formula for a triangle's area holds for all types of triangles: $\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$. Whether the triangle is acute, right-angled, or obtuse, as long as you know the base and the perpendicular height, you can use this formula. The reason is, any triangle can be considered as half of a parallelogram or constructed using rectangles and other polygons, always resulting in this relationship.

Area of Parallelogram, Rhombus, and Trapezium

For parallelograms, the area formula is $\text{base} \times \text{height}$. You can understand this by cutting and rearranging the parallelogram into a rectangle of the same base and height. For a rhombus, the area is given by $\frac{1}{2} \times (\text{product of diagonals})$. This works because a rhombus can be split into triangles and rearranged into rectangles. The area of a trapezium with parallel sides a and b and height h is $\frac{1}{2} \times h \times (a + b)$. This comes from cleverly dividing a trapezium into rectangles and triangles or pairing two trapeziums to form a parallelogram.

Finding Area of Polygons

Any polygon—be it a quadrilateral, pentagon, or hexagon—can be divided into triangles. By calculating the area of these triangles using the basic formulas, you can determine the overall area of the polygon. For quadrilaterals, drawing a diagonal divides the shape into two triangles. For hexagons, drawing lines from one vertex often helps break the shape into manageable triangles.

Practical Applications and Conversion Units

Area is important in day-to-day tasks, such as finding out how much paper is needed to cover a surface or measuring land. Standard units are used: $1 \text{ in}^2 = 6.4516 \text{ cm}^2$, $1 \text{ km}^2 = 1,000,000 \text{ m}^2$, $1 \text{ acre} = 43,560 \text{ ft}^2$. For example, an A4 sheet, with side lengths of 21 cm and 29.7 cm, has an area of $21 \times 29.7 = 623.7 \text{ cm}^2$.

Math Activities and Exploration

Hands-on activities—like dividing squares into equal parts, dissecting parallelograms, or transforming triangles into rectangles (as described in the Śulba-Sūtras)—make area concepts clearer. These exercises show the underlying relationships between different shapes and help you master area calculations through discovery and experimentation.

Important Formulas to Remember

- Area of a rectangle = length $\times$ width

- Area of a square = side $\times$ side
- Area of a triangle = $\frac{1}{2} \times$ base $\times$ height
- Area of a parallelogram = base $\times$ height
- Area of a rhombus = $\frac{1}{2} \times$ (product of diagonals)
- Area of a trapezium = $\frac{1}{2} \times$ height $\times$ (sum of parallel sides)

Sample Problems and Figure it Out Activities

Practice is essential. Attempt problems like finding missing sides when area is known, calculating area of shaded regions, finding areas involving crosspaths in rectangles, and comparing perimeters with areas. These enhance your understanding of area calculation, help with visual reasoning, and improve problem-solving skills for examinations.

General Concepts

Area answers the 'how much space' a shape covers, while perimeter measures the boundary length. The two are not directly related—shapes with the same perimeter may have different areas. Mastery of these distinctions and formulas ensures you can solve practical and theoretical problems alike.

Summary Table of Chapter Highlights

Shape	Area Formula
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Rectangle	Length $\times$ Width
Square	Side $\times$ Side
Triangle	$\frac{1}{2} \times \text{Base} \times \text{Height}$
Parallelogram	Base $\times$ Height
Rhombus	$\frac{1}{2} \times \text{Diagonal}_1 \times \text{Diagonal}_2$
Trapezium	$\frac{1}{2} \times \text{Height} \times (\text{Sum of Parallel Sides})$

These revision notes cover the core concepts of area for rectangles, squares, triangles, parallelograms, rhombuses, trapeziums, and polygons, along with practical insights and exercises that make learning math both rigorous and enjoyable.